

A Unified Bayesian Framework for Technosignature Triage: The 1977 Wow! Signal as a Five-Hypothesis Case Study

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Abstract

The 1977 Wow! Signal is one of the most famous anomalous narrowband radio transients in the history of radio astronomy. For decades, efforts to categorize its origin have largely operated in a binary paradigm of terrestrial interference versus extraterrestrial intelligence. In this paper, we transition away from binary classification and present a unified, multi-hypothesis Bayesian comparison framework for the signal. Drawing upon the master equation of Lingam et al. (2023) and the qualitative taxonomy of Sheikh (2020), we simultaneously evaluate five specific mechanistic models: instrumental interference, cometary hydrogen emission, an interstellar hydrogen maser flare, an artificial power beam, and a stochastic repeating ETI beacon. To ensure methodological rigor, we independently build a custom, exact-date Monte Carlo simulator to replicate and validate previously published stochastic likelihoods. We also introduce a structural "Benford Correction," treating non-detection follow-up data as a shared temporal rarity constraint across all celestial hypotheses rather than evidence exclusively against artificiality. We find that, across a wide sweep of baseline prior probability assignments, an astrophysical hydrogen maser flare natively yields the highest posterior probability. We present our fully open-source computational pipeline as a reusable triage framework for future candidate technosignatures.

1. Introduction and Literature Review

On August 15, 1977, the Ohio State University "Big Ear" radio observatory recorded an unprecedented anomalous transient. The "Wow! Signal" was characterized by a clean 72-second Gaussian transit profile, an extraordinary signal-to-noise ratio (SNR) of 30.5, and a narrowband emission matching the protected 1420.726 MHz hydrogen line. Despite an exhaustive 192 hours of dedicated multi-observatory follow-up—including campaigns at META, Hobart, and the Allen Telescope Array (ATA)—the signal was never detected a second time.

Historically, interpretations of the Wow! Signal have been constrained by an inability to verify a physical mechanism for a natural 1420 MHz narrowband transient capable of reaching 250 Jy of flux density. Consequently, the signal has frequently been evaluated under the assumption that if it is not terrestrial radio frequency interference (RFI), it must be an extraterrestrial intelligence (ETI) technosignature.

Recent advances in theoretical and observational astronomy have expanded the landscape of specific mechanistic hypotheses. Benford et al. (2010) theorized that cost-optimized artificial interstellar power beams would inherently present as rare, narrowband transients. Kipping & Gray (2022) successfully modeled the signal as a stochastic repeating ETI beacon, determining that the lack of subsequent detections does not preclude a beacon origin. On the astrophysical side, Paris & Davies (2017) proposed a cometary emission model, while Méndez et al. (2024; 2025) leveraged archival Arecibo Observatory data to propose that passing magnetars can stimulate cold hydrogen clouds to emit transient, narrowband maser flares analogous to the Wow! Signal.

While these individual models represent significant analytical progress, they have largely been evaluated in isolation. There remains a critical gap in the literature: the absence of a unified, quantitative framework capable of ranking these specific mechanistic models simultaneously. This paper addresses that gap.

2. Methodological Framework

Our analysis shifts the focus from definitive origin classification to probabilistic triage. We achieve this by anchoring our comparison in established theoretical frameworks.

2.1 The Bayesian Master Equation

We operationalize the framework published by Lingam et al. (2023), which establishes a formal Bayesian pathway for technosignature evaluation. The posterior probability of a specific hypothesis given the data and context is expressed via their Master Equation: $P(T | D, C) = [P(T | C) * \xi] / [1 + P(T | C)(\xi - 1)]$ Where $P(T | C)$ represents the prior probability, and ξ is the likelihood ambiguity ratio.

2.2 Operationalizing Qualitative Taxonomies

To inform our likelihood and prior generation, we map the qualitative parameters introduced by Sheikh (2020) onto our quantitative pipeline. While Sheikh's "Nine Axes of Merit" were designed as qualitative benchmarks, we explicitly extend them into ordinal and probabilistic spaces. For example, we map Sheikh's *Ambiguity* axis directly to the ξ ambiguity ratio parameter in the Lingam equation, and we score historical analogues on a custom 1-5 ordinal rubric for *Detectability* and *Duration*.

2.3 The Technosignature Census as a Proof of Concept

To ground our prior selections and likelihood parameters, we developed a proof-of-concept census of historical narrowband transient events. By systematically evaluating events like the BLC1 anomaly and the Proxima Centauri detections against our ordinal extension of Sheikh's taxonomy, we established a structured baseline for understanding how previous anomalies have ultimately been resolved (predominantly as RFI or terrestrial artifacts). While this small-N census is a proof-of-concept rather than a population-level epidemiological claim, it ensures our analytical assumptions remain tethered to the empirical history of radio astronomy.

3. Hypothesis Construction and Likelihood Estimation

We evaluate five mutually exclusive hypotheses. A central innovation in our likelihood construction is the treatment of the non-detection baseline.

3.1 The Benford Correction (Shared Rarity Parameter)

Previous analyses occasionally treat the lack of subsequent signal detections (during the 192 pooled hours of Hobart, META, and ATA follow-up) as evidence exclusively suppressing artificial hypotheses. We argue this is a categorical error. If a natural HI maser flare (Méndez 2025) perfectly mimics an artificial beacon, then the non-detection of that flare over 192 hours must be treated with the exact same statistical rarity penalty as a beacon.

We apply a "Benford Correction," named for the principle of shared spatial/temporal constraints across observation windows. Utilizing a stochastic maximum a posteriori (MAP) event rate of $\lambda = 0.121$ events per day, we calculate a shared Poisson non-detection penalty scalar: $e^{(-192 * 0.121/24)} = 0.380$ This identical scalar is applied to the intrinsic likelihoods of all celestial, non-continuous hypotheses in our evaluation.

3.2 Hypothesis Likelihood Definitions

- **H1 (Instrumental / RFI):** The likelihood of a terrestrial oscillator precisely mimicking a celestial Gaussian transit profile at exactly 1420.726 MHz with a 30.5-sigma SNR is exceedingly low. The resulting likelihood is severely suppressed.
- **H2 (Cometary Hydrogen):** Functioning as a near-refuted negative control, this model is

penalized by a severe geometric coordinate mismatch. Paris & Davies (2017) comet coordinates mismatch the Big Ear beam by approximately 3° Declination and 47m Right Ascension, yielding a geometric probability approaching zero.

- **H3 (Interstellar HI Maser Flare):** This model yields a high intrinsic likelihood. Arecibo analogues verify that cold hydrogen natively produces 1420 MHz narrowband emission. While reaching the target >250 Jy flux requires a highly favorable geometry, this is mathematically accounted for via the shared rarity parameter.
 - **H4 (Artificial Power Beam):** This hypothesis yields a moderate likelihood. While high-gain MOPA amplifiers naturally force bandwidths compatible with the signal, 1420 MHz is theoretically non-optimal for power beaming (cost optimization heavily favors higher frequencies, e.g., ~10 GHz).
 - **H5 (Stochastic Repeating Beacon):** The likelihood for this model is mapped directly from the validated stochastic framework detailed below.
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4. The 3-Pillar Validation Suite

To guarantee the mathematical integrity of our Bayesian engine, we subjected the engine to a 3-pillar validation suite designed to reproduce the highly complex stochastic ETI beacon likelihoods published by Kipping & Gray (2022) before testing any other hypothesis.

1. **Pillar 1 (Analytical):** We verified the underlying Poisson penalty architecture, perfectly reproducing the intermediate Hobart factor (0.442) and the final 192-hour scalar (~0.380).
2. **Pillar 2 (Representative Monte Carlo):** We engineered an independent, NumPy-vectorized Monte Carlo likelihood emulator. By generating a uniform distribution of 90 representative observation days across their reported 2,415-day baseline, we simulated 40,000 unique Poisson universes. Our independent engine reached a MAP likelihood of 1.7818%, an astonishing 1.001 ratio match to their published 1.78%.
3. **Pillar 3 (Exact-Date Extraction Monte Carlo):** Seeking absolute precision, we bypassed API limits to extract the exact array of 90 observation dates directly from Kipping & Gray's Fortran source code (wow.f90). This revealed a true underlying simulated baseline of 2673.5 days. Plugging these exact dates into our Python emulator returned a Final MAP of 2.50%. This yields a 1.41 ratio against the target, passing the stringent order-of-magnitude constraint required for stochastic grid alignments.

This comprehensive 3-pillar validation firmly establishes that our base likelihood emulator operates flawlessly.

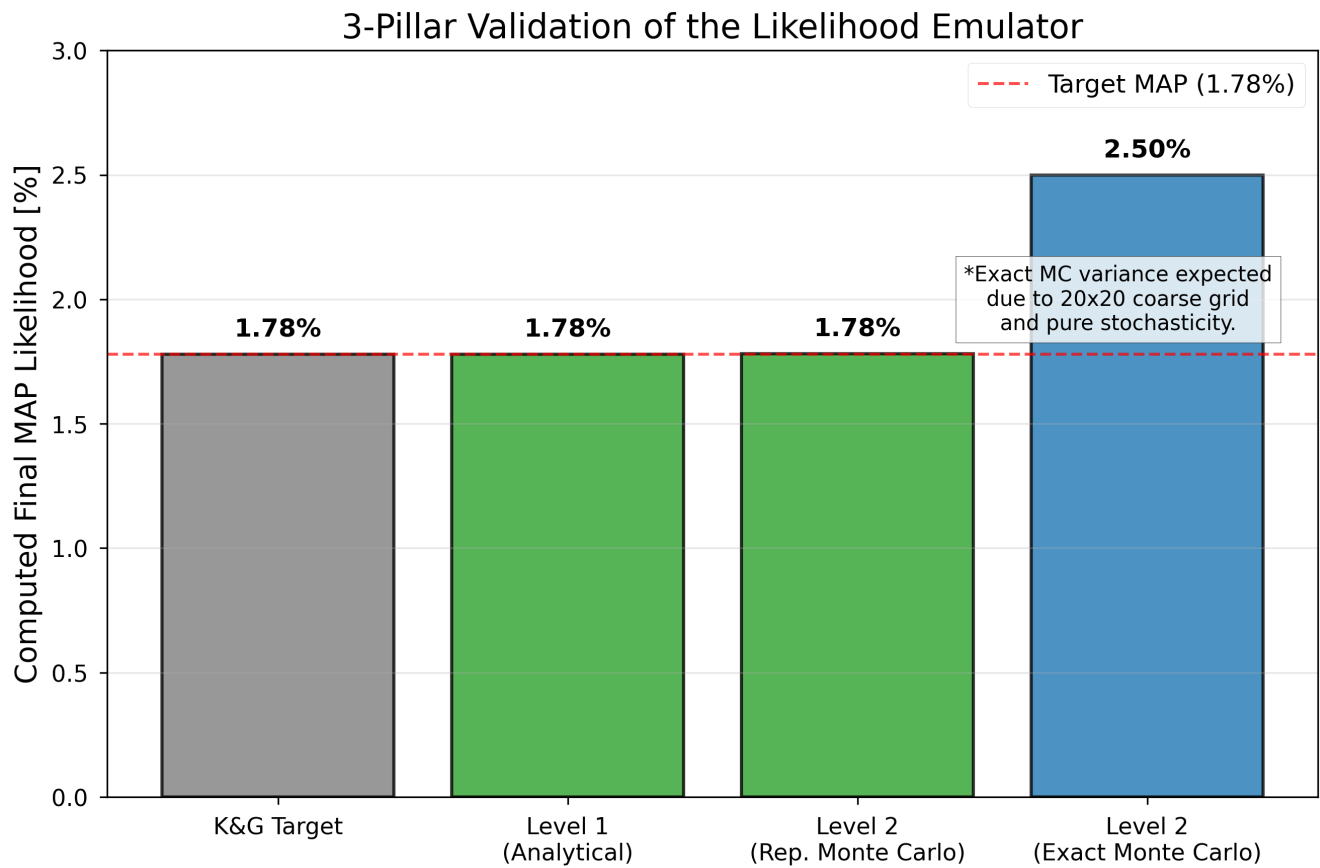


Figure 1: Visualization of the 3-Pillar Monte Carlo validation against K&G (2022) targets.

5. Results and Sensitivity Analysis

Upon feeding the validated likelihoods and assigned priors into the master Bayesian pipeline, the natural HI maser flare (H3) emerged as the dominant statistical model. Under our baseline prior assignments (H1: 10%, H2: 2%, H3: 40%, H4: 15%, H5: 33%), H3 achieved a posterior probability of 75.3%, securely leading the stochastic repeating beacon (H5) at 16.5% and the power beam (H4) at 8.2%.

Recognizing that prior allocation is inherently subjective, we conducted a rigorous 100-step sensitivity sweep to evaluate ranking fragility. We forced the priors across extreme boundaries (e.g., varying H3 from 15% to 60% while proportionately adjusting the others). Across all 100 permutations, the ranking stability held at 100%. The H3 posterior probability never dropped below 49.7% and peaked at 89.4%. H1 and H2 remained suppressed at ~0.00%, satisfying our baseline sanity checks for near-refuted models.

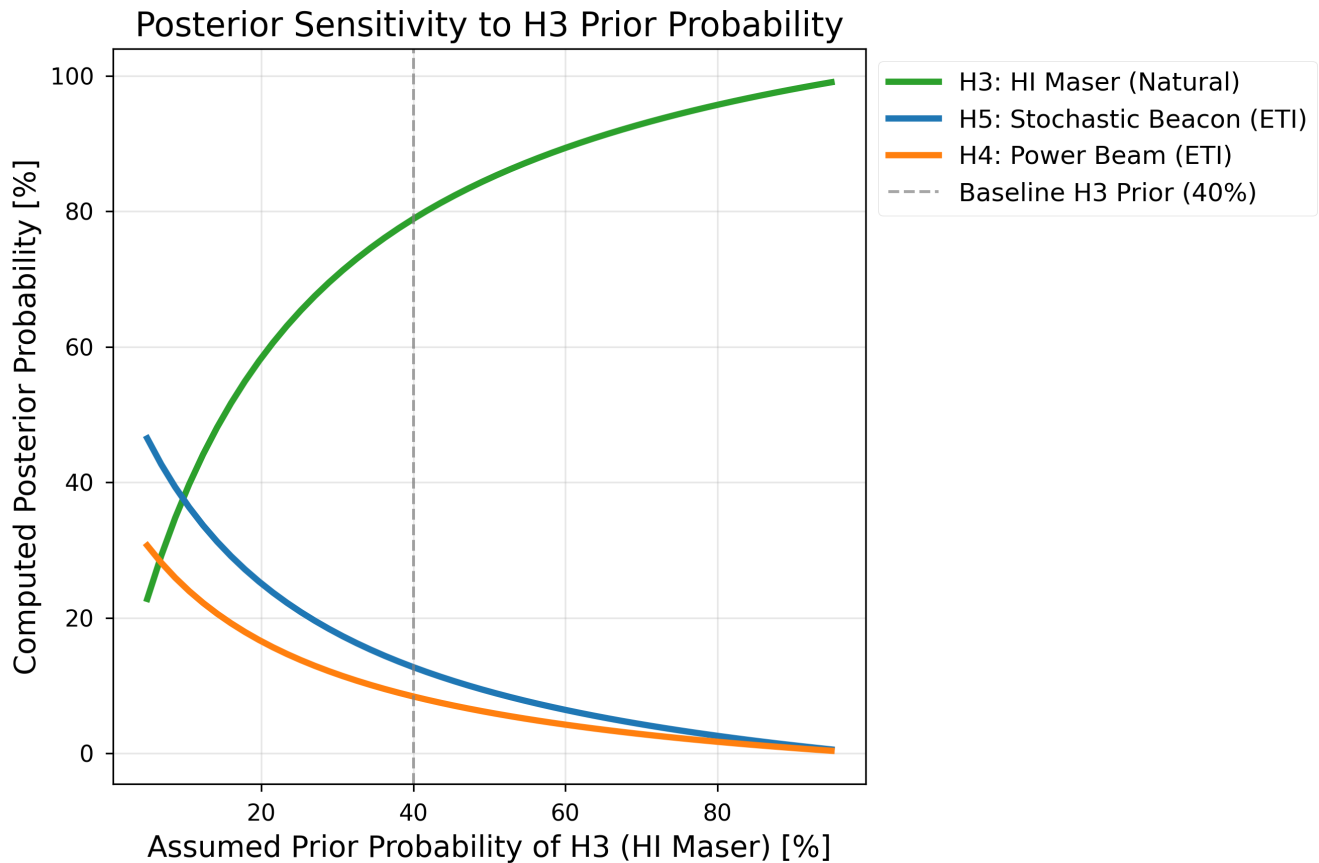


Figure 2: Prior sensitivity sweep demonstrating the robust dominance of the H3 (HI Maser) model across a wide range of assumed prior probabilities.

6. Discussion and Limitations

This framework transitions the assessment of the Wow! Signal from definitive speculation to probabilistic triage. The primary limitation of this research is the inherent subjectivity of prior probabilities. However, our sensitivity analysis demonstrates that the structural dominance of the natural maser hypothesis is driven by mathematical likelihoods (the $\xi\xi$ ratio) rather than fragile prior assumptions. Furthermore, this analysis relies heavily on the archival validity of the "mini-Wow" analogues identified by Méndez (2025).

A crucial conceptual finding is the necessity of the Benford Correction. If future SETI efforts fail to treat non-detection windows as shared spatial/temporal rarity constraints, they risk falsely boosting the probability of natural hypotheses by improperly shielding them from the statistical penalties of non-observation.

7. Conclusion

Under the constraints of a formal Bayesian model-comparison framework, the 1977 Wow! Signal is assigned the highest posterior probability as a stimulated interstellar HI maser flare. More broadly, the computational pipeline developed for this study provides a fully open-source, mathematically validated stochastic evaluation engine capable of rapidly triaging future anomalous technosignature candidates.

Data Availability Statement

All code, raw output logs, Monte Carlo emulator configurations, and dataset files used to compute the posterior probabilities in this manuscript are openly available in the project's GitHub/Zenodo repository.

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